
“My Nanny, Dudu”
Jacques Henri Lartigue
(C. 1902)

Light By Light

Theoretical Developments
in Gravitational Wave Physics

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1919. May 29 eclips confirms that gravity “bends” light


REVOLUTION IN SCIENCE.

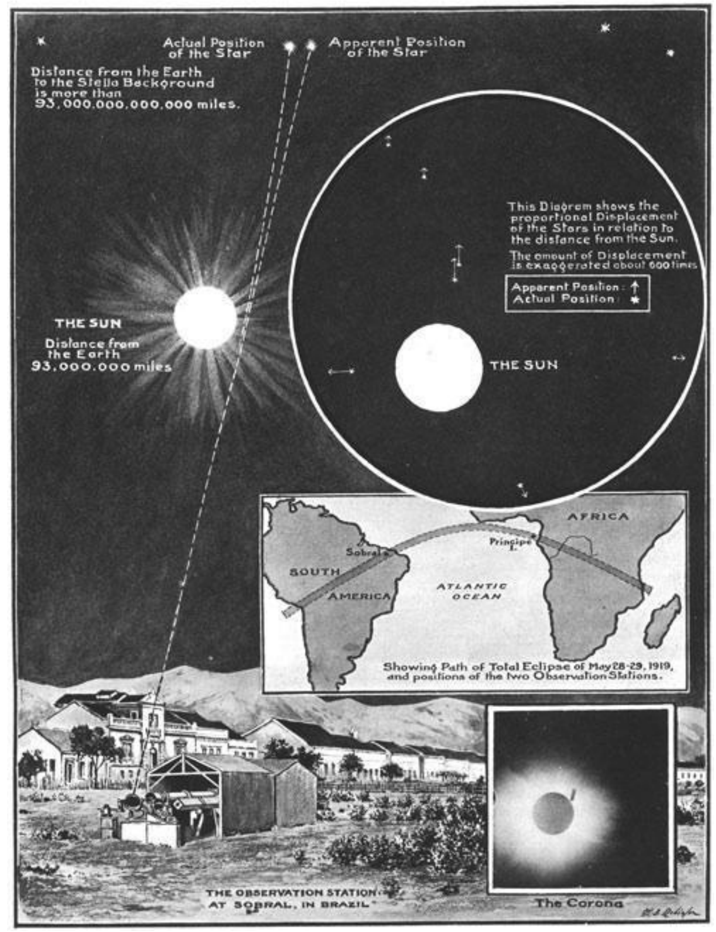
NEW THEORY OF THE UNIVERSE.

NEWTONIAN IDEAS OVERTHROWN.

Yesterday afternoon in the rooms of the Royal Society, at a joint session of the Royal and Astronomical Societies, the results obtained by British observers of the total solar eclipse of May 29 were discussed.

The greatest possible interest had been aroused in scientific circles by the hope that rival theories of a fundamental physical problem would be put to the test, and there was a very large attendance of astronomers and physicists. It was generally accepted that the observations were decisive in the verifying of the prediction of the famous physicist, Einstein, stated by the President of the Royal Society as being the most remarkable scientific event since the discovery of the predicted existence of the planet Neptune. But there was differ-

‘Times of London‘, Nov 7 1919



'Illustrated London News‘, Nov 22 1919

Kinetic energy dominated processes: plan

Hoop Conjecture & end of short distance physics

Universal limit on maximum luminosity?

Relativistic head-on collisions of particles and black holes

Threshold collisions

Particles at the light ring

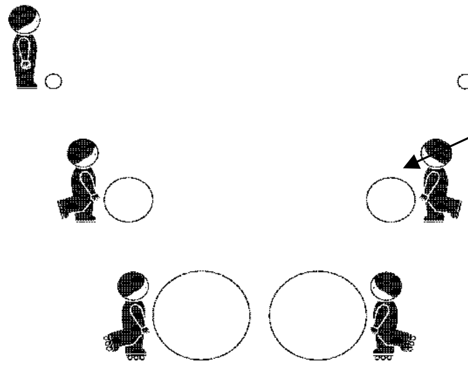
Colliding pulses of radiation

Hoop Conjecture and kinetic-dominated processes

(Thorne 1972; Flanagan 1983; Schoen & Yau 1983)

"An imploding object forms a BH when, and only when, a circular hoop with circumference 2π the Schwarzschild radius of the object can be made that encloses the object in all directions."

Large amount of energy in small region



This is the hoop

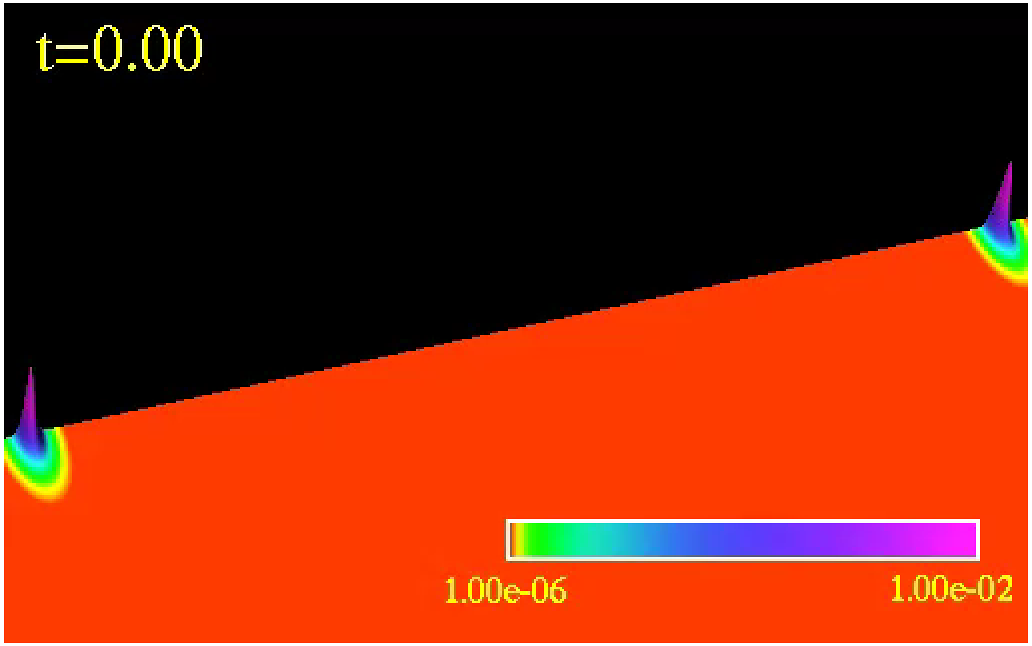
$R=2G\gamma M/c^2$

Size of electron: 10^(-17) cm

Schwarzschild radius: 10^(-55) cm

The end of short-distance physics

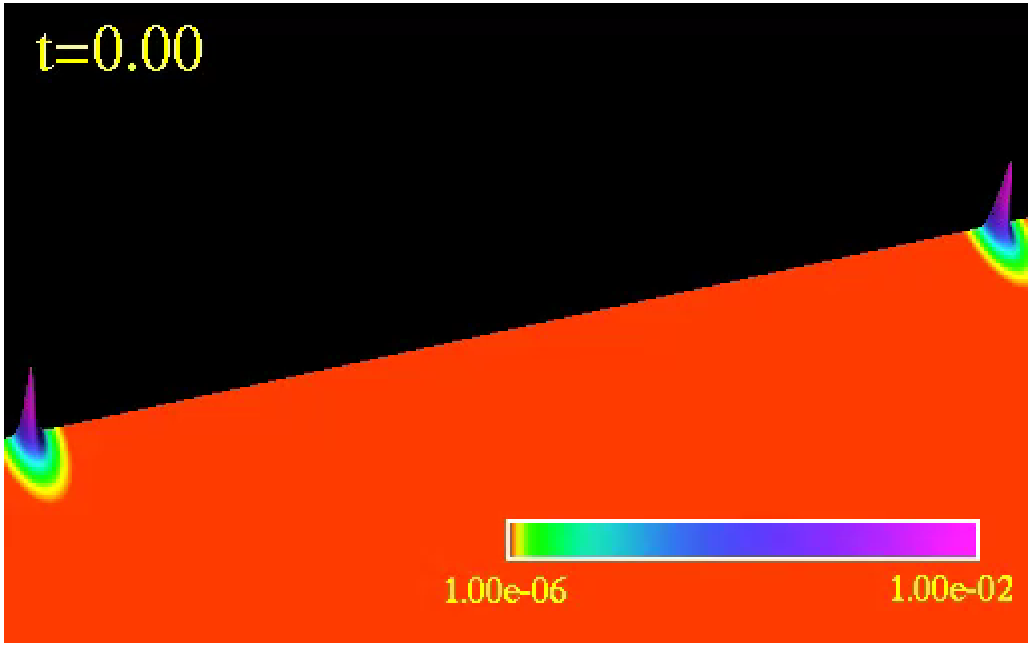
Lorentz boost = 2



Choptuik & Pretorius, Phys.Rev.Lett. 104:111101 (2010)

The end of short-distance physics

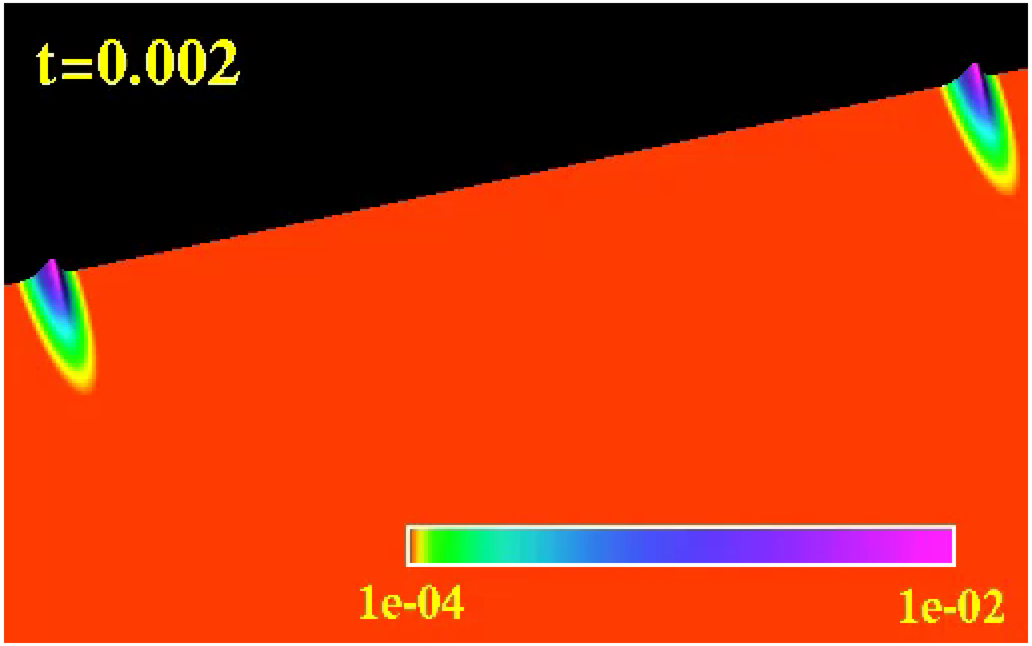
Lorentz boost = 2



Choptuik & Pretorius, Phys.Rev.Lett. 104:111101 (2010)

The end of short-distance physics

Lorentz boost = 4

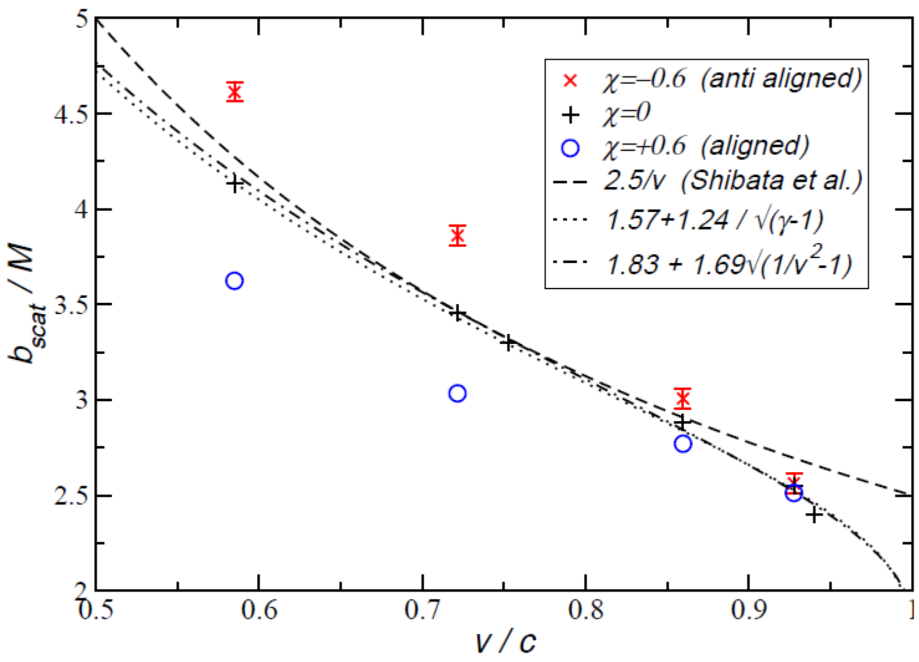


Choptuik & Pretorius, Phys.Rev.Lett. 104:111101 (2010)

Black holes do form in high energy collisions

(Choptuik & Pretorius 2010; East & Pretorius 2013; Rezzolla & Takami 2013; Exposito-Farino+ 2026)

Structure of colliding objects is irrelevant



Sperhake + PRL111:041101 (2013)

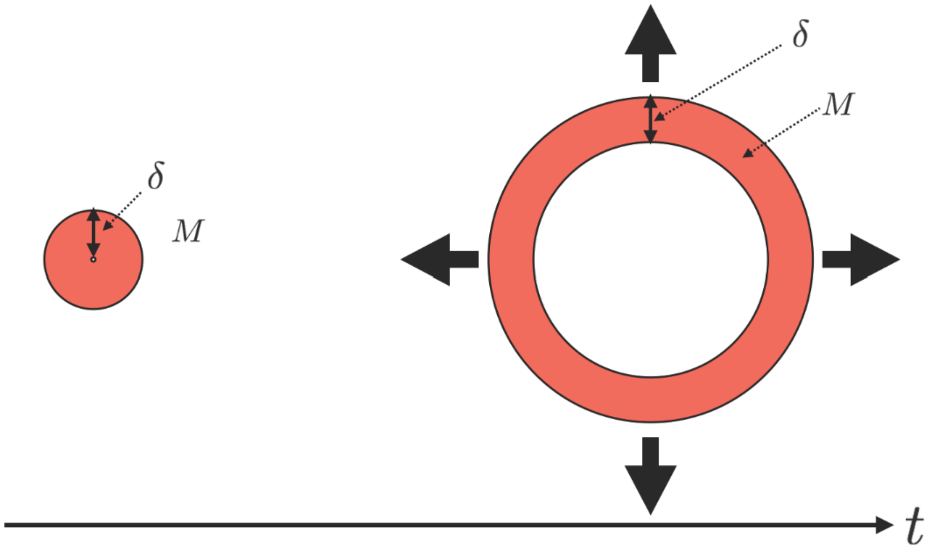
On the maximum luminosity

Event	Peak Luminosity
Three Gorges dam	$3\times10^{-43}\,\mathcal{L}_{\rm P}$
Most powerful laser	$3\times10^{-38}\,\mathcal{L}_{\rm P}$
Tsar Bomba	$3\times10^{-27}\,\mathcal{L}_{\rm P}$
Solar luminosity	$1\times10^{-30}\,\mathcal{L}_{\rm P}$
γ-ray bursts	$1\times10^{-5}\,\mathcal{L}_{\rm P}$
Inspiralling BHs	$2\times10^{-3}\,\mathcal{L}_{\rm P}$
High-energy BH collision	$2\times10^{-2}\,\mathcal{L}_{\rm P}$
Critical collapse	$2\times10^{-1}\,\mathcal{L}_{\rm P}$
End point of BH evaporation	$\mathcal{L}_{\rm P}$

$$\mathcal{L}_{\rm P} = \frac{c^5}{G} = 3.6 \times 10^{52}\,\mathrm{W}$$

Gibbons & Barrow MNRAS 446:3874 (2015); Cardoso+ PRD97:084013 (2018)

The Thorne-Dyson conjecture



$$\text{luminosity} = \frac{Mc^2}{\delta/c} < \frac{c^5}{G}$$

K. Thorne, Gravitational Radiation (North-Holland 1983)

F. J. Dyson, Interstellar Communication (Benjamin, NY, 1963)

Gibbons & Barrow MNRAS 446:3874 (2015);

Cardoso+ PRD97:084013 (2018)

High energy collisions

“End of short distance physics”

Cosmic Censorship, black hole formation and critical phenomena

Possible long-lived metastable states

If kinetic-energy dominated, how much can be radiated?

Universal limit on maximum luminosity c^5/G (10^59 erg/sec)?

Can multiple black holes form by focusing radiation?

How do pencils of light interact gravitationally?

How does light interact with gravitational waves?

Consequences for light ring physics, for physics of lasers

Zero Frequency Limit

(Weinberg '64; Smarr '77)

Take two free particles, changing abruptly at t=0

$$T^{\mu\nu} = \sum_{i=1,2} \frac{P_i^\mu P_i^\nu}{E_i} \delta^3(x - v_i t)\theta(-t) + \frac{P_i'^\mu P_i'^\nu}{E_i'} \delta^3(x - v_i' t)\theta(t)$$

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,, \qquad h_{\mu\nu} = 4 \int \frac{T_{\mu\nu}(t_r, x') - 1/2\eta_{\mu\nu}(t_r, x')}{|x' - x|} d^3x'$$

$$\frac{dE}{d\omega d\Omega} = \frac{M^2\gamma^2 v^4}{\pi^2} \frac{\sin^4\theta}{(1 - v^2\cos^2\theta)^2}$$

Radiation isotropic in the UR limit, multipole structure $E_l \propto \frac{1}{l^2}$

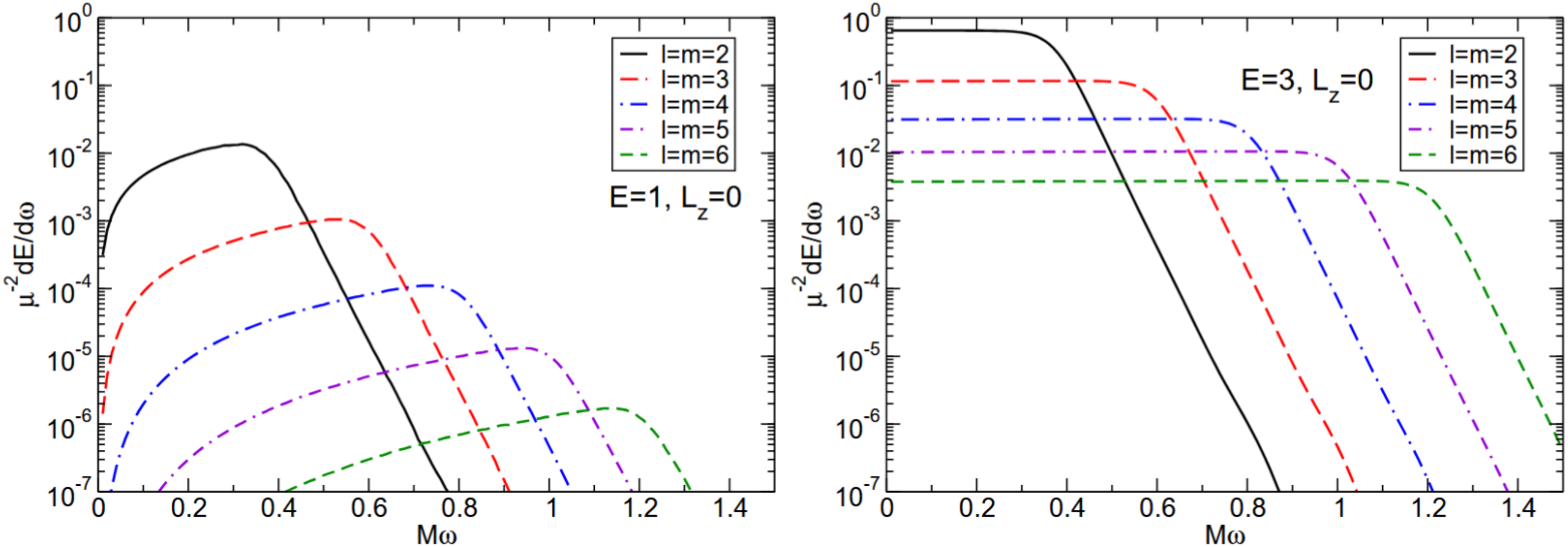
Functional relation $E_{rad}(\gamma)$, flat spectrum

Roughly 65% of maximum possible at γ=3

With cutoff M $\omega\approx$0.4 we get 25% efficiency for conversion of gws

Point particles colliding with black holes

(Davis+ 1971; Cardoso & Lemos 2002, Berti+ 2010)

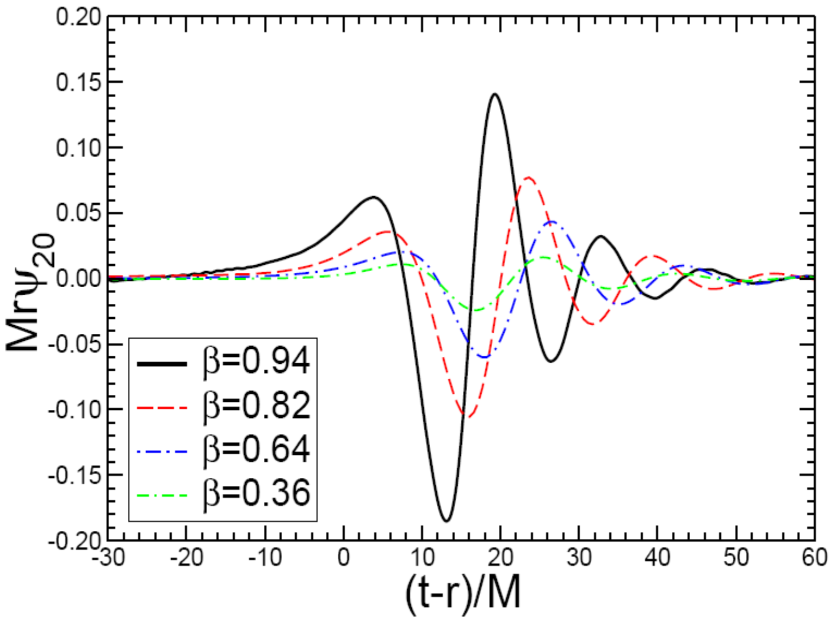


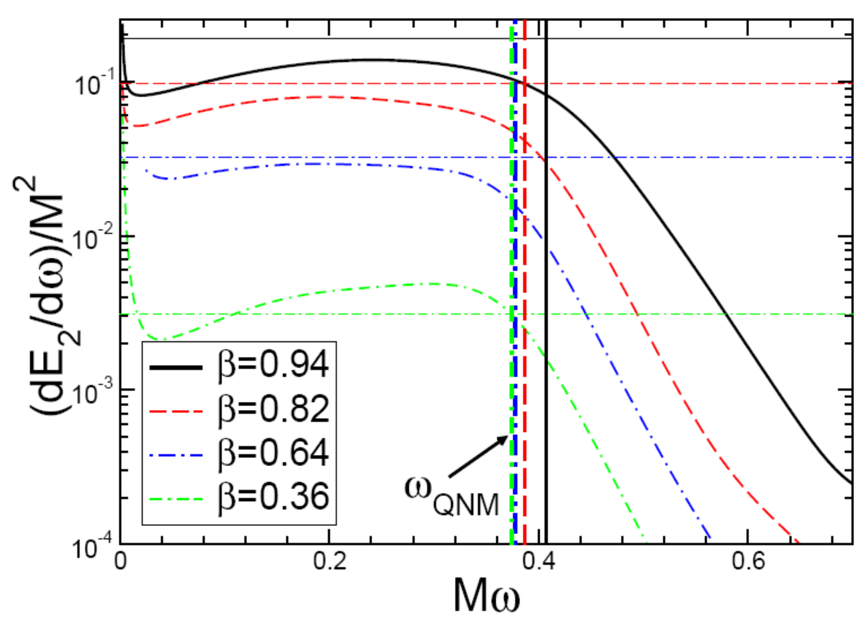
Radiation isotropic in the UR limit, multipole structure $E_l \propto \frac{1}{l^2}$

Functional relation $E_{rad}(\gamma)$, flat spectrum

Roughly 65% of maximum possible at γ=3

With cutoff $\omega = \omega_{QNM}$ we get 13% efficiency for conversion of gws





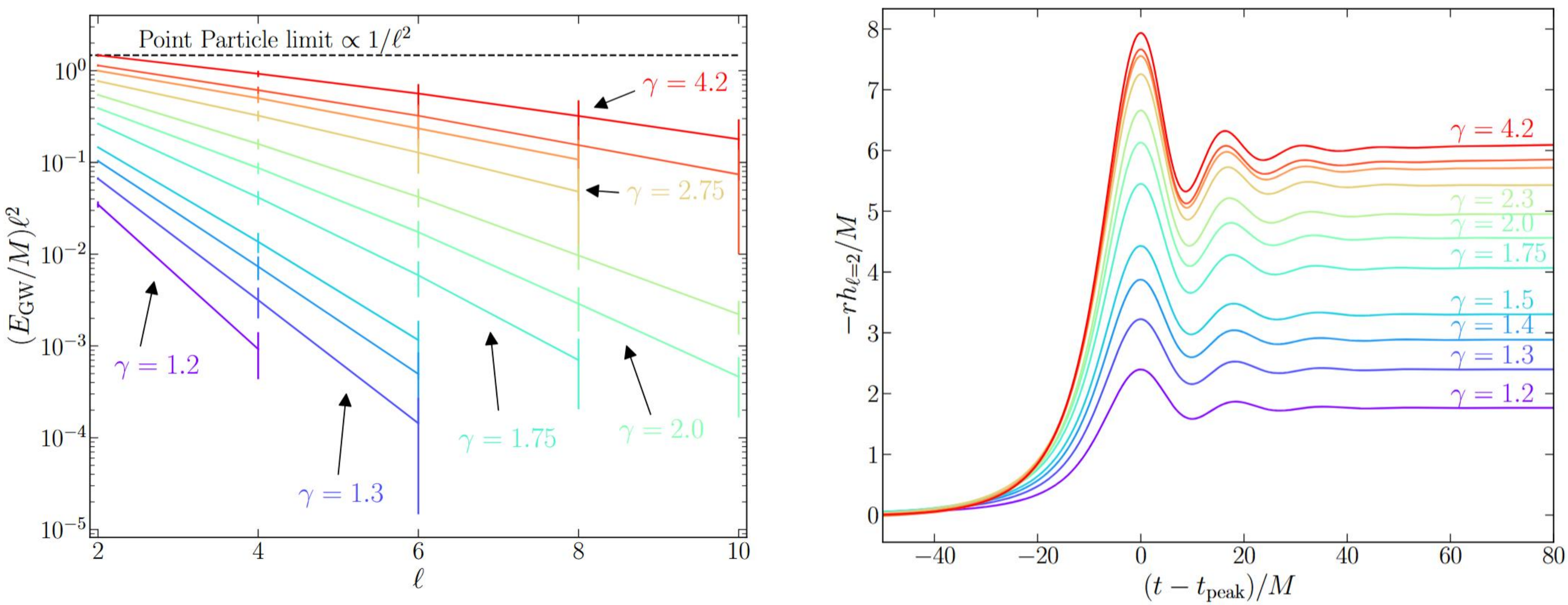
Waveform is almost just ringdown

Spectrum is flat, in good agreement with ZFL

Substantial memory, as predicted by ZFL

Cutoff frequency at the lowest quasinormal frequency

Sperhake + PRL101:161101 (2008)



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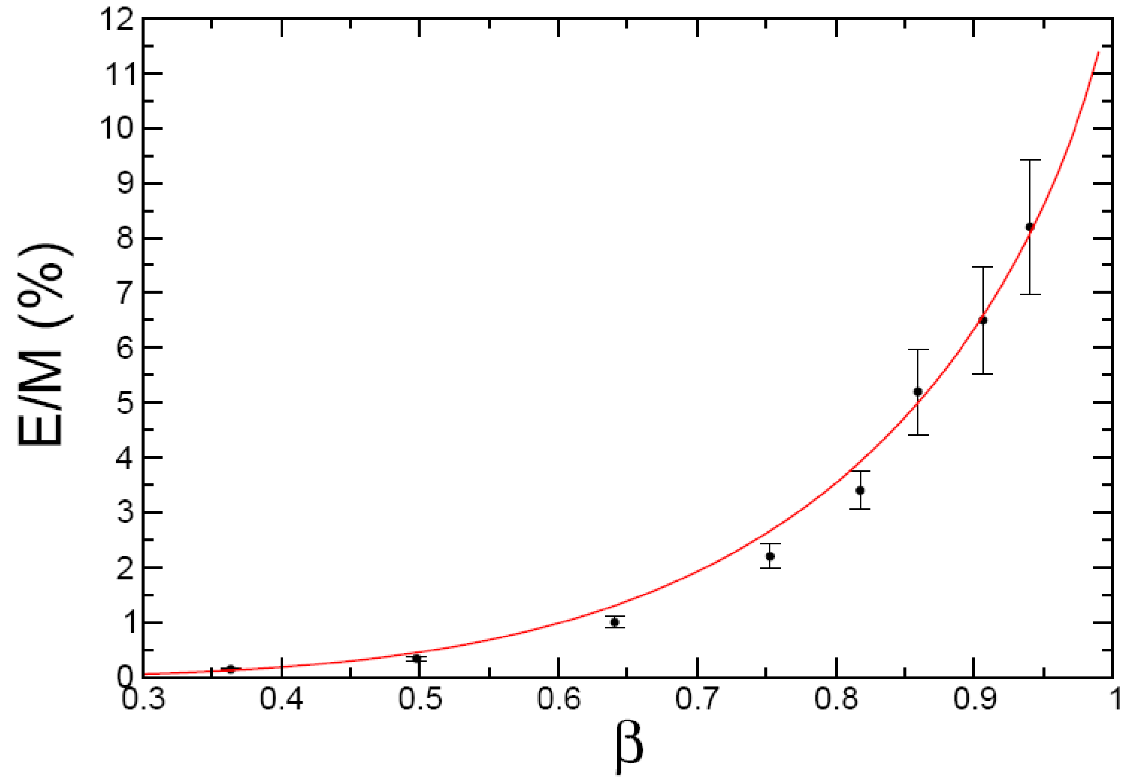
Sperhake + PRL101:161101 (2008); Helfer + (unpublished)



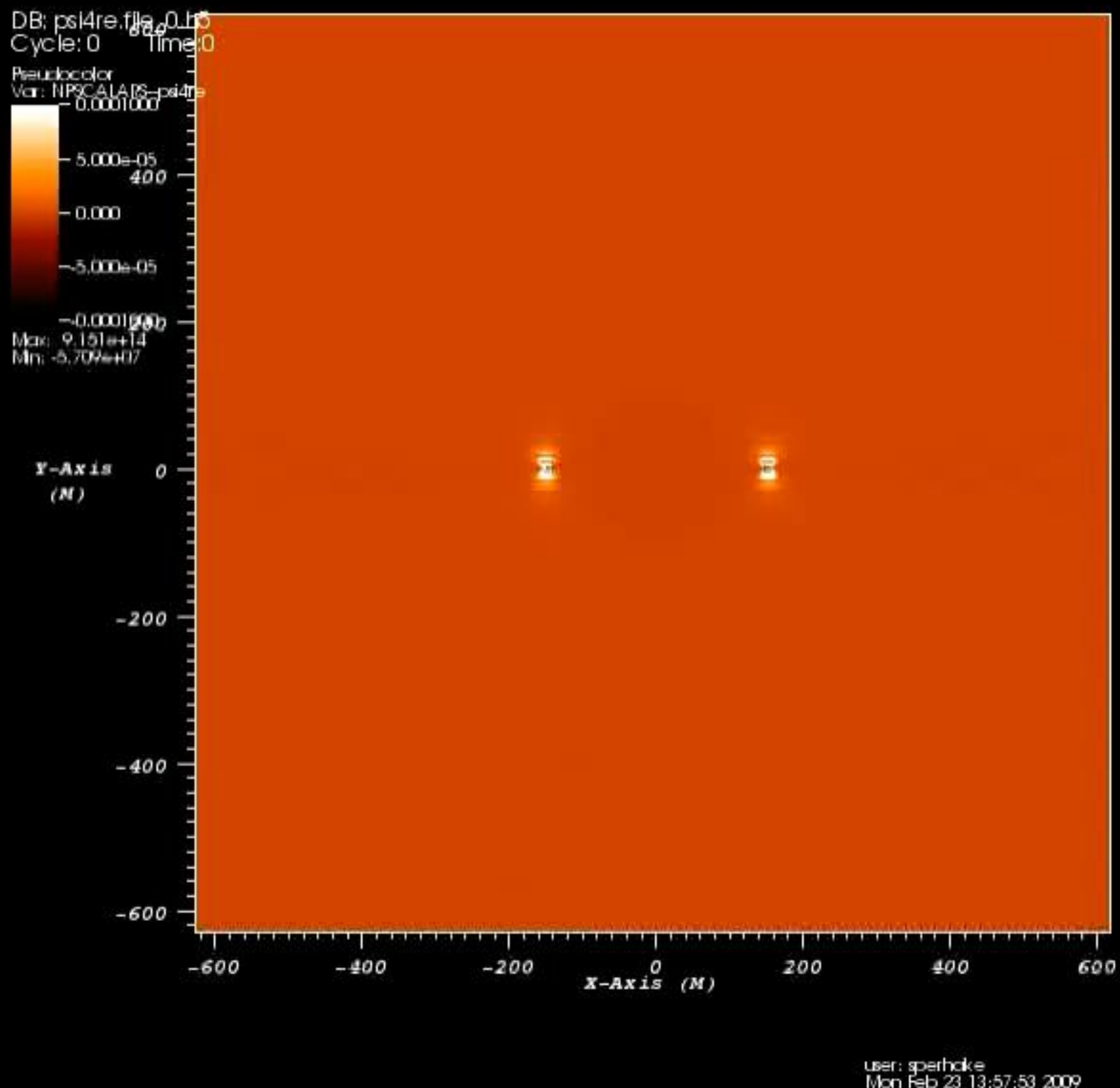
13%

$$\frac{E}{M} = E_{\infty}\left(\frac{1+2\gamma^2}{2\gamma^2} + \frac{(1-4\gamma^2)\log\left(\gamma+\sqrt{\gamma^2-1}\right)}{2\gamma^3\sqrt{\gamma^2-1}}\right)$$

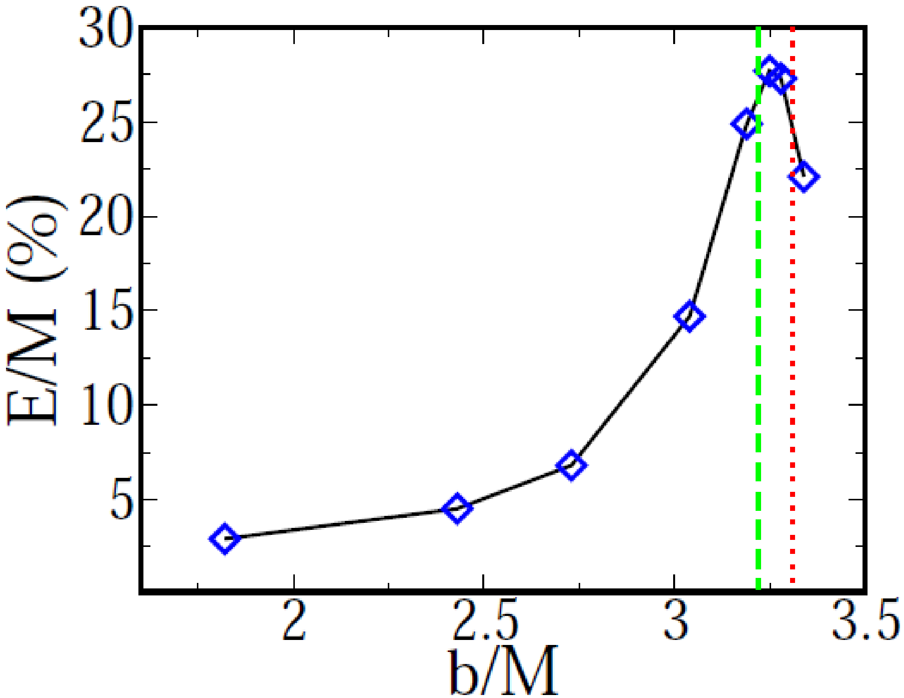
Cosmic Censorship: can we destroy black holes?

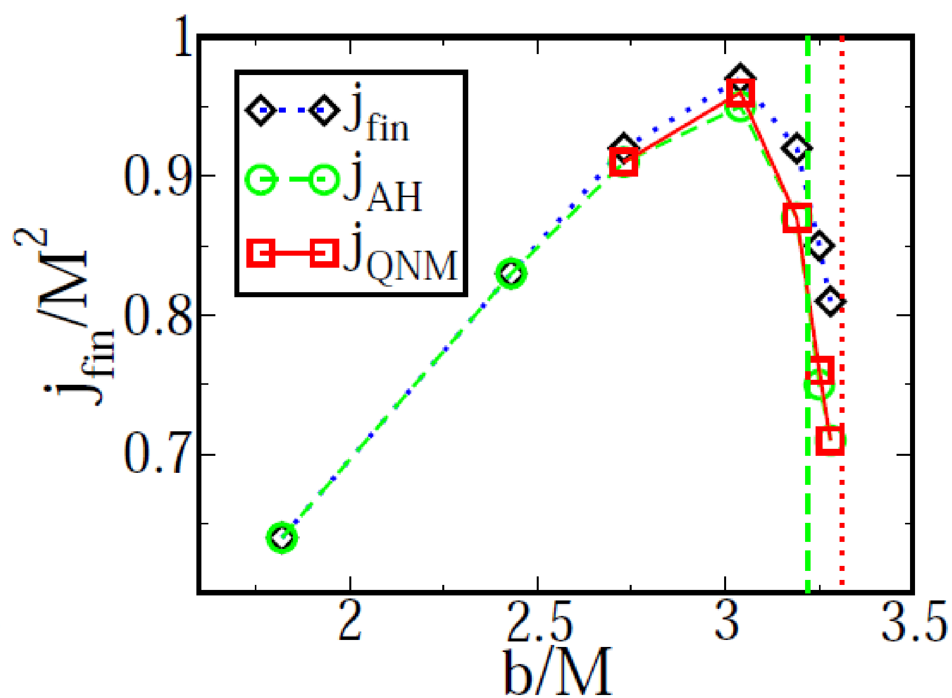
$$\frac{cJ}{GM^2} \leq 1$$

Black holes have small "rotation"



Sperhake + PRL103:131102 (2009)





More than 25% (35%) CM energy radiated for v=0.75 c (0.92c)!

Final BH rapidly spinning. CC is preserved

Conjecture 50% of radiation as in point particle

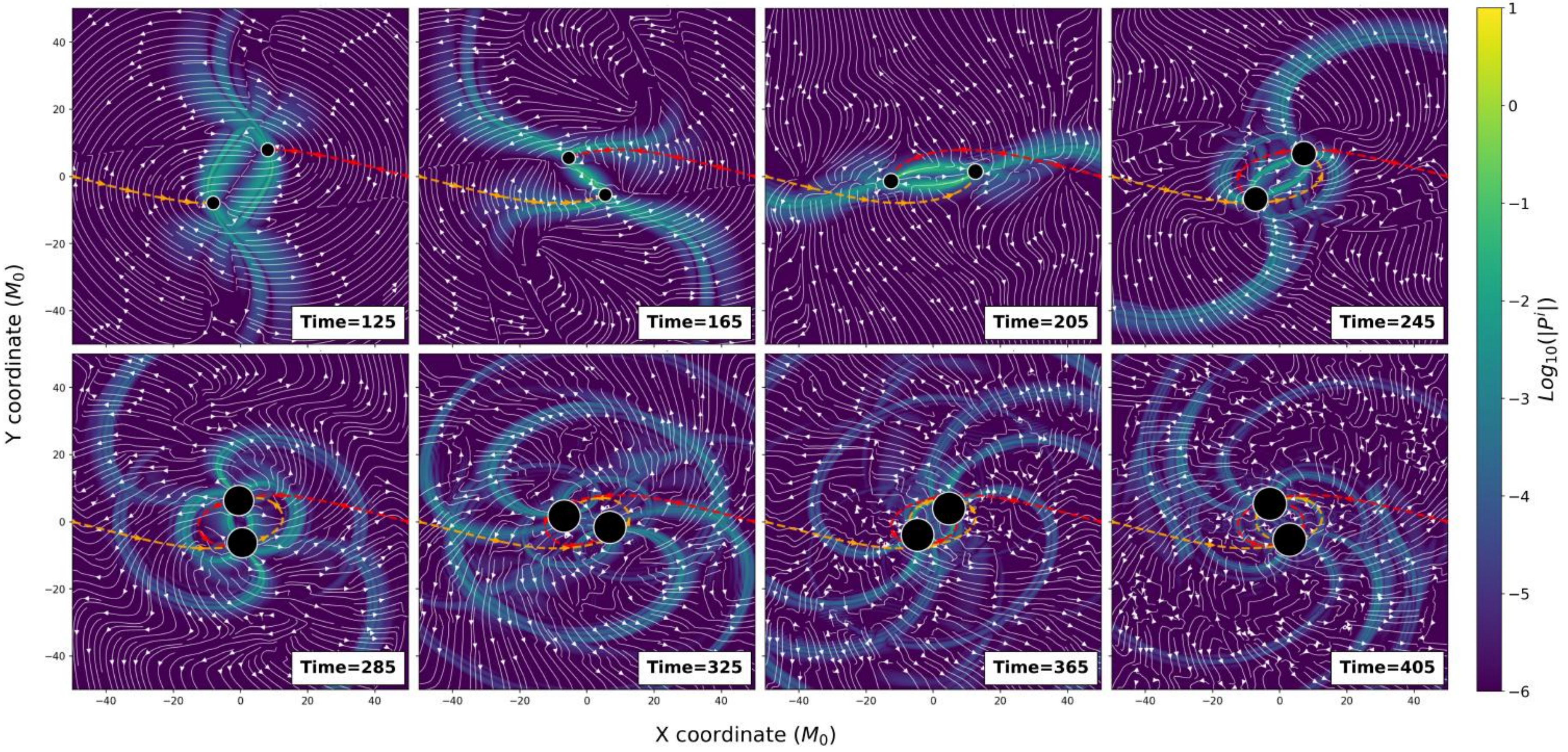


FIG. 3. Snapshots of the Bel–Robinson super-Poynting flux $|\boldsymbol{P}|$ on the orbital plane at different times (in units of M_0) for a near-threshold encounter with $b = 2.46M_{\mathrm{ADM}}$. (See Fig. 1 for the corresponding waveform at large radius.) The dashed curves mark the past trajectories of the black holes (circles), with the circle sizes scaled to represent the instantaneous irreducible masses of the black holes as measured from their apparent horizons [56, 57]. Horizon properties as a function of evolution time for this encounter are detailed in Fig. 4. During the early approach, each boosted hole carries a strongly Lorentz-contracted near-zone curvature pattern. The "curvy" nature of the wave fronts in the initial frame is in part a gauge effect. Following the initial close encounter, this curvature flux is focused, sheared, and repeatedly lensed within the interaction region, eventually producing the intricate flux pattern that corresponds to the delayed pulses seen in Fig. 1. At late times, both horizons have grown substantially through absorption, and the black holes fly apart with a mildly relativistic speed of $\sim 0.13c$.

Zhu + arXiv:2604.26253; see also Gundlach + PRD 86, 084022 (2012)

Light orbiting black holes

(Barausse+ 2021)

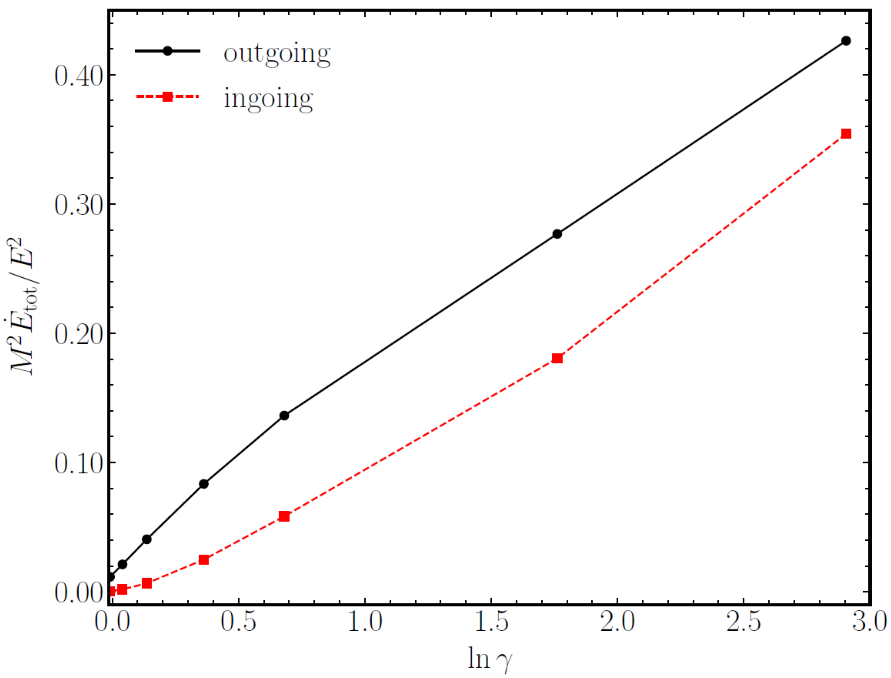


FIG. 2. Total instantaneous gravitational-energy fluxes $\dot{E}_{\rm tot}$ of a point particle around a Schwarzschild BH as a function of the relativistic boost factor $\gamma \equiv E/m_p$. We show the gravitational energy flux normalized by the square of the Killing energy E^2, both at infinity and at the horizon.

Light orbiting black holes

(Barausse+ 2021)

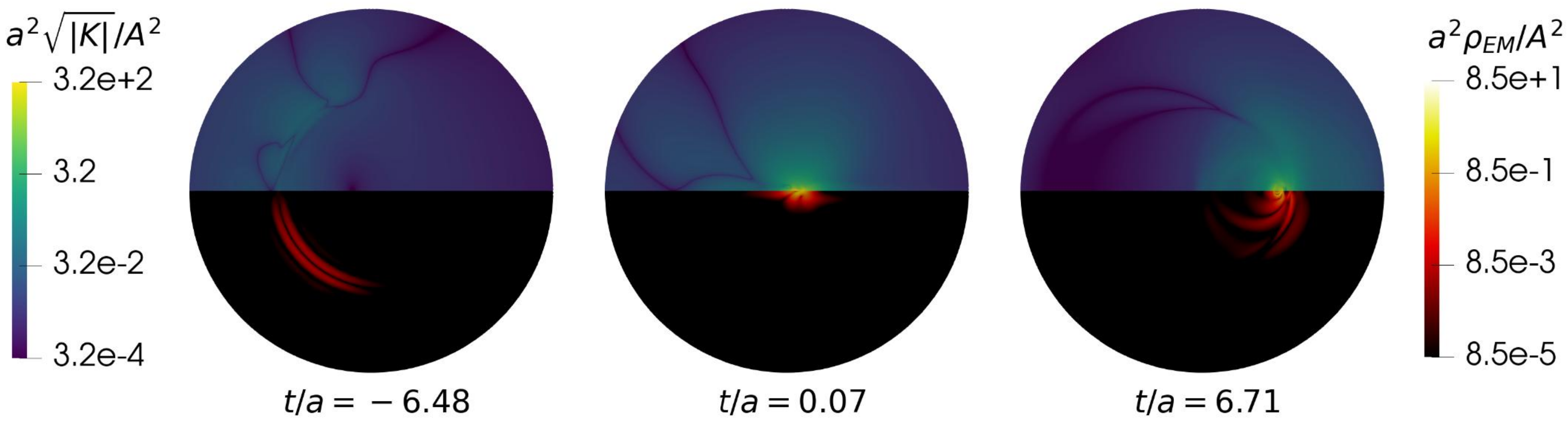
$$
\begin{aligned}
\dot{E} &= \frac{c_0\gamma^2}{\ell} e^{-c_1\ell/\gamma^2} \frac{m_p^2}{M^2} \\
r &= 3M\left(1+\frac{1}{9\gamma^2}\right), \\
c_0 &= 0.07,\ c_1 = 0.42 \qquad \text{in and out fluxes}
\end{aligned}
$$

At light ring

$$
\begin{aligned}
\dot{E} &= 0.064 \frac{E^2}{M^2\ell} \\
\ell_{\text{cutoff}} &= \frac{\lambda}{2\pi r} \\
\frac{\delta E}{E} &\sim 10^{-78} \qquad \text{per cycle for SgrA}
\end{aligned}
$$

Collisions of pulses of radiation

see also Redondo-Yuste's talk for nonlinear content of gravity



$$
\begin{aligned}
\Pi_k &= \tilde{c} A \delta_k^z \mathrm{Re}\left[\frac{a^5(a+it)}{3}\frac{3\bar{\rho}^4 - 6z^2\bar{\rho}^2 - z^4 + 8iz\bar{\rho}^3}{\bar{\rho}^3(\bar{\rho}^2+z^2)^3}\right] \\
\bar{\rho}^2 &= \rho^2 + (a+it)^2 = x^2 + y^2 + (a+it)^2 \\
(\tilde{\varphi}, \tilde{A}^i) &= (0, \epsilon^{ijk}\partial_j \Pi_k)\,.
\end{aligned}
$$

Lekner, Proc.R.Soc.Loondon (2018); Exposito-Patino + (ongoing)

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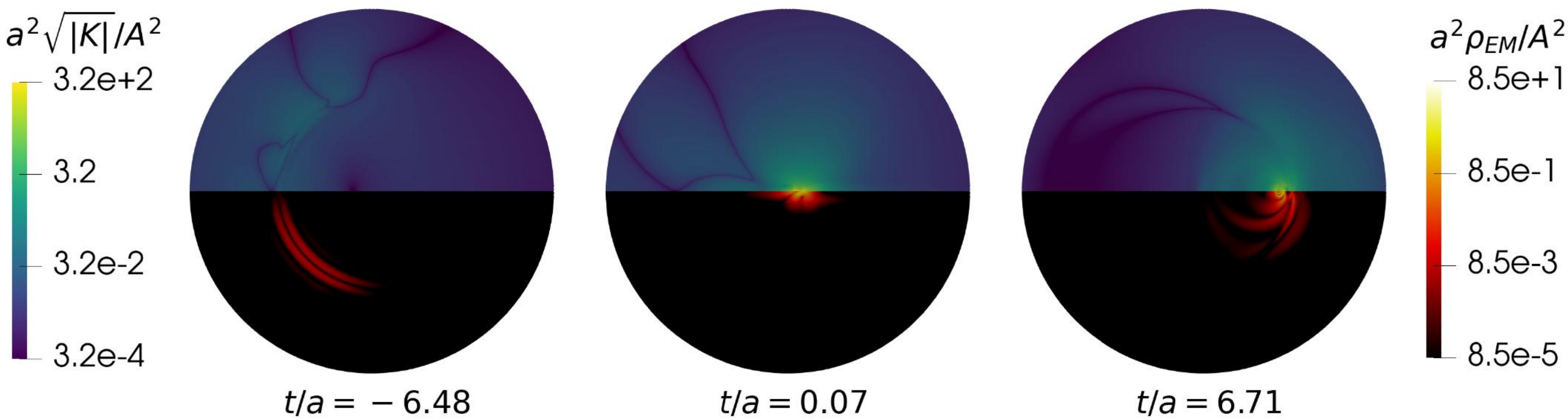
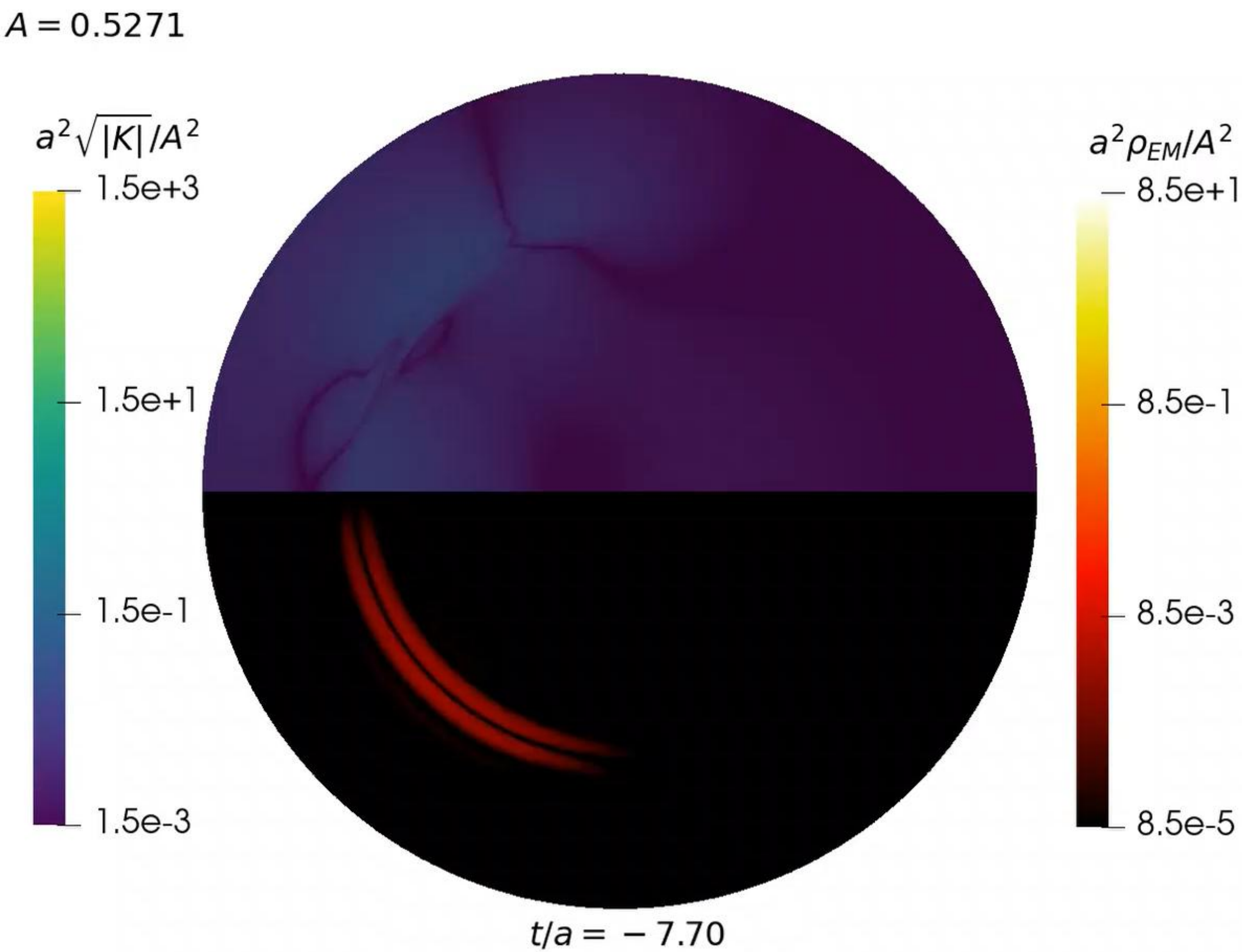


$$E = 10^{27}\,\frac{1\,\mathrm{m}}{a}\,\mathrm{V/m}$$

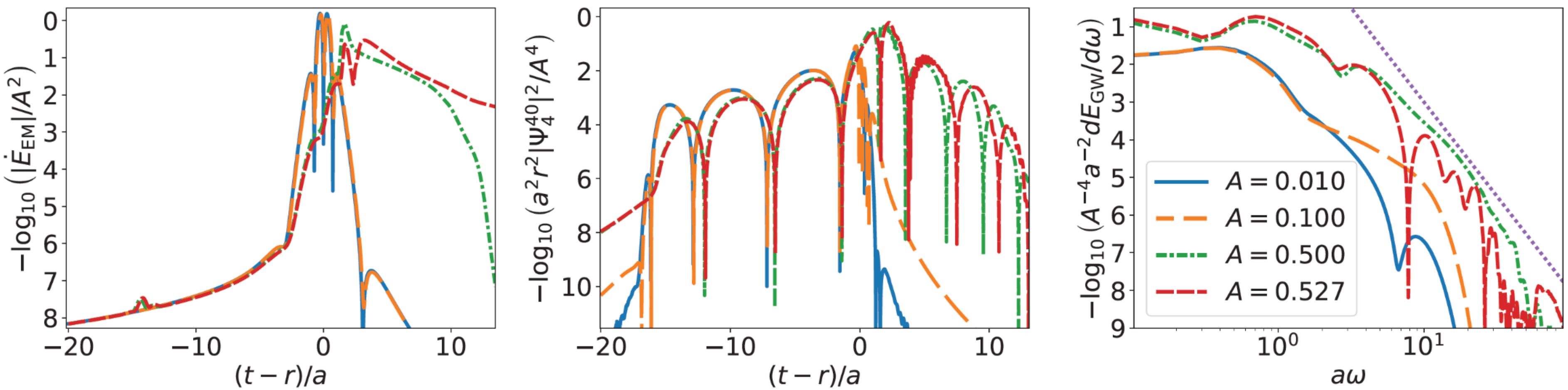
Lekner, Proc.R.Soc.Loondon (2018); Exposito-Patino + (ongoing)

The bearable lightness of light



Exposito-Patino + (ongoing)

The weight of light



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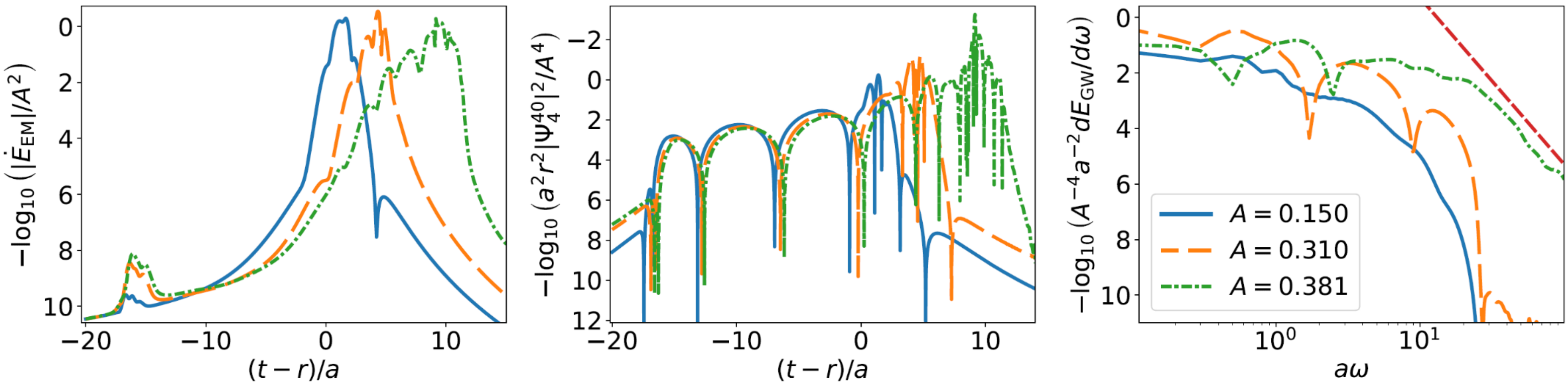

Exposito-Patino + (ongoing)

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